1-40kw-g1 首航光伏逆变器

ModBus-RTU 通信协议

2013-8-17

Version Record

Setting Time	Version	Modify content	Operator
20180320	V1.00	Title Change,Initialization control	资志翔
20190617	V1.01	Compare with the code of 3-6K & 20-33K, change parts the error codes; delete parts of the communication protocol(like battery); add some model content of Fuse	石应杰

Summary of Agreement

1.1 Physical Layer

Physical Transmission: RS485/RS422
Address： 1 ～ 63
Baud rate: 9600bps
Maxi mun Distance: 1000M
Medium: Shielded twisted pair (STP)
Mode: MODBUS — RTU

1.2 Link Layer

Transmission: Master-slave + Half duplex

First, The Master addressing the terminal device(Slave), then on the opposite direction, Slave transmit the response to the Master. The protocol only used for data transmit between Master and Slave.Data Transmit through independent devices are prohibit. So that the data from independent device will not occupy the communication channel during the initialization, it is limit to signal inquiry request to the Master

Data Transmission Format: 1 starting byte; 8 data bytes; 1 stop byte; Parity: none
Data Transmission Format:

Slave Address	Function Code	Data	CRC
1-Byte	1-Byte	N-Byte	2byte

The Protocol defines CRC,DATA and other other related parameters , these are necessary for specific data exchange. When data frame arrive at the slave, it will addressing to its specific device,then problematic head will be remove , the data will read by the device.If there are no errors, the slave will perform the tasks request by the data, the slave will have data back to the sender, by add problematic head and send the data back to the sender, the response data send back from the slave contains the following contents: The terminal slave address(Address), the executed commands(Function),data generated after executed the requested command(Data) and a parity check code(Check), The Salve can identify the wrong message from the Master, and make different response.

1.3（ Address ）

Address is the beginning of the frame,compose by one byte(value 1-63). This address indicates the identity of the device that specified by user. This slave will get the data from the connected Master. For whole system, Address for each slaves must be unique. The slave will feedback when it addressed, when slave back with response,The Slave address data will show which slave connect with Master.

1.4（ Function ）

Function show the operation that slave will take after addressed. All Function code & meaning support by monitor device list in Table 2

Function Code	Range of Register Address	Meaning
0x03	0x0000-0x00FF Read the register data from inverter input	Get one or more values from Register
	0x0100-0x01FF Read the register data from built- in combiner' input	Get one or more values from Register
0x04	0x1000-0x10FF Read the register data keep by inverter	Get one or more values from Register
	0x1100-0x11FF Read the register data keep by built-in combiner	Get one or more values from Register
0x13	0x1000-0x10FF Write Setting or parameters to inverter	Offer one or more values from Register
	0x1100-0x11FF Write setting or parameter to built-in combiner	Offer one or more values from Register
0x21	Ext Code 0x2000-0x20ff Inverter	
	Ext Code 0x2100-0x21ff Bulit-in combine	
0x07	Conceal Function 0x3000 Calibration(with password?)	
0x08	Conceal Function 0x4000 Maintaining information	
0x02	Auto Timing	
0x01	Remote on/off control, power limit ; power factor	
0x50	Read Storage data from EEPROM	
0x51	Write in EEPROM storage data	
0x61	Read the data from SD	
0x10	Read access time	
0x30	Factory Reset	
0x31	Clear today's energy	
0x32	Recovery the fault setting of current country code setting	
0x33	Clear the total energy	
0x34	Clear the event list	
0x35	Read the control character of relay	

0x36	write the control character to Relay	
0x37	Read the control character of relay, set the fault ID, WHEN WE get alarm for control character(configable)	
0x38	Set the fault ID (alarm, can configable) as the control character to relay	
0x45	Read the testing Flag	
0x46	Write in the test Flag	

1.5 （Data）

Data Consists by two different hexadecimal number systems, range 00H～FFH. It made up of RTU character according to the network transmission mode. The data from Master to Slave need contains the additional information: Slave must execute the behavior defined by function code.(in-consecutive register address included); number of the pending items need to be process; actual number of data bytes in domain. For example:Function Code to Slave,read a register, data need indicate start and number of specific register, embedded address data types, different slave will have different result, because the features of slaves are different.

1.6（CRC）

The verification allow the error exist during the transmission between Master and Slave. Sometimes because of the electrical noise and other interruptions, a set of data may change during the transmission from one device to another. The CRC verification ensure that Master or Slave not respond to the incorrect data that has changed during the transmission. This verification improve the security and efficiency of whole system.We are using 16-bit circulation Redundary Check(CRC16),CRC occupies two bytes,contained one 16-bit Binary value.CRC value calculated by transmitting device,then attach to the data frame, the receiving device recalculate the CRC value . Compare with the received CRC value, if two value are not equal, an error arise.

Set all bytes to “1” for a 16-bit register,Then operation by 8 bit byte in frame and current value of register continuity, only 8 data bits per byte participate in generation CRC value.start bits; stop bits and other possible parity bits will not affect CRC value generation.During the CRC value generation, each 8-bit byte XOR with the content in register,then shift the result to low-bit, high-bit supplement with “0”， the last significant bit(LSB) remove and test. If the LSB is “1”, the register XOR with a preset fixed value, if the LSB show”0”, not do any treatment. Repeat the above process until perform 8 times shift operation. The next 8-bit bytes XOR with current value of register, also perform another 8 shift XOR operation as above -mentioned, the final value we get is the CRC value.

The Process of generating a CRC value
Step1: Preset all bytes to “1” For 16-bit register, defined as CRC register(0FFFFH)
Step2: XOR the first 8-bit byte in data frame with the low byte in CRC register
Step3: Shift the CRC register one bit to the right,high-bit supplement with “0”, the LSB remove and test
Step4: If the LSB is “0”, repeat step3(Next Shift), if the LSB is “1”, the register XOR with a preset fixed value, （0A001H）
Step 5:Repeat step 3 &4, until perform 8 times shift operation.

Step 6: Repeat from step2 to 5 to deal with next 8-bit byte, until end of all bytes processing complete.
Step 7: Finally， the value of CRC register is the value of CRC

Instructions

2.1 Broadcast data frame information (address 0x88)

Broadcast data with no response

2.1.1. Auto Timing

(Slave Address)	0x88
(Function Code)	0x02
Register Address(Hi)	0x50
Register Address(Lo)	0x00
Number of Registers(Hi)	0x00
Number of Registers(Lo)	0x03
Data Field(Second)	
Data Field(Minute)	
Data Field(Hour)	
Data Field(Date)	
Data Field(Month)	
Data Field(Year)	
CRC16 Lo	
CRC16 Hi	

Address Table(Auto Timing)

Address	Definition	Variable Type	Length	Range	Fault Value	Remarks
0x5000	Auto Timing	BCD				

2.1.2 ON/OFF SINGAL

1. Remote ON/OFF Control

(Slave Address)	0x88
(Function Code)	0x01
Register Address(Hi)	0x01
Register Address(Lo)	0x42
Value of Registers(Hi)	0x00
Value of Registers(Lo)	0x55/0x66
CRC16 Lo	0x82
CRC16 Hi	0xBB

Remarks : Inverter ON Register Lo=0x55
Inverter Off Register Lo=0x66

2.1.3 Active power Derated Setting

1. Active power Derated

(Slave Address)	0x88
(Function Code)	0x01
Register Address(Hi)	0x01
Register Address(Lo)	0x41
Value Number of Register(Hi)	
Value of Register(Lo)	
CRC16 Lo	
CRC16 Hi	

2.1.4 Power factor Setting(Reactive power)

1. Power factor Setting(Reactive power)

(Slave Address)	0x88
(Function Code)	0x01
Register Address(Hi)	0x01
Register Address(Lo)	0x61
Value of Register(Hi)	
Valuer of Register(Lo)	
CRC16 Lo	
CRC16 Hi	

2.1.5 Reactive Power Setting

1. Reactive power setting

(Slave Address)	0x88
(Function Code)	0x01
Register Address(Hi)	0x01
Register Address(Lo)	0x62
Value of Register(Hi)	
Valuer of Register(Lo)	
CRC16 Lo	
CRC16 Hi	

2.2 Read Command 1（0x03）

Through the 0x03 function code,queries allow the register information,data format is as follows:

2.2.1 Read Data Format

Master request message Format :

Slave Address	Function Code	Starting Address	Number of Registers	CRC16
1 Byte	1 Byte	1 Byte	1 Byte	1 Byte
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Lo Byte Hi Byte

Slave response message Format：

Slave Address	Function Code	Byte Count	Register-1 value	...	Regiser-N value	CRC16
1 Byte	1 Byte	1 Byte	1 Byte	N-2	1 Byte	1 Byte
Byte	Byte	Byte	Hi Byte Lo Byte	...	Hi Byte Lo Byte	Lo Byte Hi Byte

Example(query the state of the Inverter)

Query:

Slave Address	0x01
Function Code	0x03
Register Address Hi	0x00
Register Address Lo	0x00
Number of Registers Hi	0x00
Number of Registers Lo	0x01
CRC16 Lo	0x84
CRC16 Hi	0x0A

Response:

Slave Address	0x01
Function Code	0x03
Byte Count	0x02
Value of Register Hi	0x00
Value of Register Lo	0x00
CRC16 Lo	0xB8
CRC16 Hi	0x44

2.2.2 Read inverter data through Address List

Operating state

00： wait

01：check

02：Normal

03：Fault

04：Permanent

Fault Message:

Byte0

bit	Error Message	Detailed(ID code)
Bit0	GridOVP	Grid Over Voltage Protection ID01
Bit1	GridUVP	Grid Under Voltage Protection ID02
Bit2	GridOFP	Grid Over Frequency Protection ID03
Bit3	GridUFP	Grid Under Frequency Protection ID04
Bit4	PVUVP	PV Under Voltage Protection ID05
Bit5	GridLVRT	Grid Low Voltage Ride Protection ID06
Bit6		7
Bit7		8

Byte1

bit	Error Message	Detailed(ID code)
Bit0	PVOVP	PV Over Voltage Protection ID09
Bit1	IpvUnbalance	PV Input Current Unbalance ID10
Bit2	PvConfigSetWrong	PV Input Mode Configure Wrong ID11
Bit3	GFCIFault	Ground-Fault Circuit Interrupters Fault ID12
Bit4	PhaseSequenceFault	Phase Sequence Fault ID13
Bit5	HwBoostOCP	Hardware Boost Over Current Protection ID14
Bit6	HwAcOCP	Hardware AC Over Current Protection ID15
Bit7	AcRmsOCP	The Grid Current is too high ID16

Byte2

bit	Error Message	Detailde(ID code)
Bit0	HwADFaultIGrid	The Grid Current Sampling Fault ID17
Bit1	HwADFaultIGrid	The DCI Sampling Fault ID18
Bit2	HwADFaultVGrid	The Grid Voltage Sampling Fault ID19
Bit3	GFCIDeviceFault	GFCI Device Sampling Fault ID20
Bit4	MChip_Fault	Main Chip Fault ID21
Bit5	HwAuxPowerFault	Hardware Auxiliary Power Fault ID22
Bit6	BusVoltZeroFault	BUS Voltage Zero Fault ID23

Bit7	IacRmsUnbalance	The unbalance output current ID24

Byte3

bit	Error Message	Detailde(ID code)
Bit0	BusUVP	Bus Under Voltage Protection ID25
Bit1	BusOVP	Bus Over Voltage Protection ID26
Bit2	VbusUnbalance	Bus Voltage Unbalance ID27
Bit3	DciOCP	DCI Over Current Protection ID28
Bit4	SwOCPInstant	The Grid Current is too high ID29
Bit5	SwBOCPInstant	The Input Current is too high ID30(Software protection)
Bit6	reserved	31
Bit7	reserved	32

Byte4，byte5

bit	Error Message	Detailed(ID code)
Bit0	reserved	33/41
Bit1	reserved	34 /42
Bit2	reserved	35/43
Bit3	reserved	36/44
Bit4	reserved	37/45
Bit5	reserved	38/46
Bit6	reserved	39/47
Bit7	reserved	40/48

Byte6

bit	Error Message	Detailed(ID code)
Bit0	ConsistentFault_VGrid	The Grid voltage sampling consistency error between the master and slave DSP ID49
Bit1	ConsistentFault_FGrid	The Grid frequency sampling consistency error between the master and slave DSP ID50
Bit2	ConsistentFault_DCI	The DCI sampling consistency error between the master and slave DSP ID51
Bit3	ConsistentFault_GFCI	The GFCI sampling consistency error between the master and slave DSP ID52
Bit4	SpiCommLose	The communication Fail between the master and slave DSP ID53
Bit5	SciCommLose	The communication Fail

		between the slave and communication board ID54
Bit6	RelayTestFail	The relay Fault ID55
Bit7	PvIsoFault	Low insulation resistance between the PV array and earth ID56

Byte7

bit	Error Message	Detailed(ID code)
Bit0	OverTempFault_Inv	The inverter temp is too high ID57
Bit1	OverTempFault_Boost	The boost temp is too high ID58
Bit2	OverTempFault_Env	The environment temp is too high ID59
Bit3	PEConnectFault	No PE wire connect for inverter ID60
Bit4	reserved	61
Bit5	reserved	62
Bit6	reserved	63
Bit7	reserved	64

Byte8

bit	Error Message	Detailed(ID code)
Bit0	unrecoverHwAcOCP	The Grid current is too high,and cause unrecoverable fault ID65
Bit1	unrecoverBusOVP	The bus voltage is too high,and cause unrecoverable fault ID66
Bit2	unrecoverIacRmsUnbalance	The Grid current is unbalance,and cause unrecoverable fault ID67
Bit3	unrecoverIpvUnbalance	The unbalance input current cause unrecoverable fault ID68
Bit4	unrecoverVbusUnbalance	The unbalance bus voltage cause unrecoverable fault ID69
Bit5	unrecoverOCPInstant	The Grid current is too high,and cause unrecoverable fault ID70
Bit6	unrecoverPvConfigSetWrong	PV Input Mode configure wrong,and cause unrecoverable fault ID71
Bit7	reserved	72

Byte9

bit	Error Message	Detailed(ID code)
Bit0	reserved	73
Bit1	unrecoverIPVInstant	The input current is too high,and cause unrecoverable fault ID74

Bit2	unrecoverWRITEEEPROM	The EEPROM Fault ID75
Bit3	unrecoverREADEEPROM	The EEPROM Fault ID76
Bit4	unrecoverRelayFail	The relay Fault cause unrecoverable fault ID77
Bit5	reserved	78
Bit6	reserved	79
Bit7	reserved	80

Inverter alert message:byte0

bit	Error Message	Detailed(ID code)
Bit0	OverTempDerating	The inverter derated because of the high temperatureID81
Bit1	OverFreqDerating	The inverter derated because of the High Grid frequency ID82
Bit2	RemoteDerating	The inverter derated by remote control ID83
Bit3	RemoteOff	The inverter has shut down by remote control ID84
Bit4	UnderFreqDerate	The inverter has derated because of the lower Grid frequency ID 85
Bit5	reserved	Reserved ID86
Bit6	reserved	Reserved ID87
Bit7	RefluxDerating	The inverter has derated because of the reflux power ID88

Inverter alert message: byte1

bit	Error Message	Detailed(ID code)
Bit0	reserved	reserved
Bit1	reserved	reserved
Bit2	reserved	reserved
Bit3	reserved	reserved
Bit4	reserved	reserved
Bit5	reserved	reserved
Bit6	reserved	reserved
Bit7	reserved	reserved

Communication board inner message:byte0

bit	Error Message	Detailed(ID code)
Bit0	Fan1 alarm	Fan 1 alarm ID91
Bit1	Fan2 alarm	Fan 2 alarm ID92
Bit2	Lightning protection alarm	Lightning protection alarm ID93
Bit3	Software version is not consistent	Software version is not consistent ID94
Bit4	Communication board EEPROM fault	The communication board EEPROM is fault ID95

Bit5	RTC clock chip anomaly	RTC clock chip is fault ID96
Bit6	InValidCountry	The country code is invalid ID97
Bit7	SDfault	The SD card is fault ID98

Communication board inner message:byte1

bit	Error Message	Detailed(ID code)
Bit0	Fan3 alarm	Fan 3 alarm ID90
Bit1	reserved	reserved
Bit2	reserved	reserved
Bit3	reserved	reserved
Bit4	reserved	reserved
Bit5	reserved	reserved
Bit6	reserved	reserved
Bit7	reserved	reserved

Inverter Data Address table

Address	Definition	Variable type	Length	Range	Default value	Remarks
0x0000	Operating state	Uint	16			Only Low-Byte availability
0x0001	Fault 1	Uint	16			High-Byte:byte1 Low-Byte:byte0
0x0002	Fault 2	Uint	16			High-Byte:byte3 Low-Byte:byte2
0x0003	Fault 3	Uint	16			High-Byte:byte5 High-Byte:byte4
0x0004	Fault 4	Uint	16			High-Byte:byte7 Low-Byte:byte6
0x0005	Fault 5	Uint	16			High-Byte:byte9 Low-Byte:byte8
PV Input Message						
Address	Definition	Variable type	Length	Range	Default value	Remarks
0x0006	PV1 voltage	Uint	16	0-1000V		Unit:0.1V

0x0007	PV1 current	Uint	16	0-100A		Unit:0.01A
0x0008	PV2 voltage	Uint	16	0-1000V		Unit:0.1V
0x0009	PV2 current	Uint	16	0-100A		Unit:0.01A
0x000A	PV1 power	Uint	16	0-100kw		Unit:0.01kw
0x000B	PV2 power	Uint	16	0-100kw		Unit:0.01kw
Output Grid Message						
Address	Definition	Variable type	Length	Range	Default value	Remarks
0x000C	Active power(output)	Uint	16	0-1000V		Unit:0.01kw
0x000D	Reactive power(output)	Uint	16	0-1000V		Unit:0.01kVar
0x000E	Grid frequency	Uint	16	0-1000V		Unit:0.01Hz
0x000F	A-phase voltage	Uint	16	0-1000V		Unit:0.1V
0x0010	A-phase current	Uint	16	0-1000V		Unit:0.01A
0x0011	B-phase voltage	Uint	16	0-1000V		Unit:0.1V
0x0012	B-phase current	Uint	16	0-1000V		Unit:0.01A
0x0013	C-phase voltage	Uint	16	0-1000V		Unit:0.1V
0x0014	C-phase current	Uint	16	0-1000V		Unit:0.01A
Inverter Generation Message						
Address	Definition	Variable type	Length	Range	Default value	Remarks
0x0015	Total production high-byte	Uint	16	0-65536		Unit:1kWh
0x0016	Total production low-byte	Uint	16	0-65536		
0x0017	Total generation time high-byte	Uint	16	0-65536		Unit:1 hour
0x0018	Total generation time low-byte	Uint	16	0-65536		
0x0019	The day generation	Uint	16	0-1000V		Unit:0.01kWh
0x001A	The day generation time	Uint	16	0-65536		Unit:1 minute
Inverter Inner Message						
Address	Definition	Variable type	Length	Range	Default value	Remarks
0x001B	Inverter module temperature	Uint	16	0-1000V		Unit: 1℃
0x001C	Inverter inner temperature	Uint	16	0-1000V		Unit: 1℃
0x001D	Inverter Bus voltage	Uint	16	0-1000V		Unit:0.1V
0x001E	PV1 inout voltage sampling by slave CPU	Uint	16	0-1000V		Unit:0.1V

0x001F	PV2 input voltage sampling by slave CPU	Uint	16	0-1000V		Unit:0.1V
0x0020	Count-down time	Uint	16			
0x0021	Inverter alert message	Uint	16			
0x0022	Input mode					0x00:iparallal 0x01: independent
0x0023	Inverter inner message					
0x0024	Insulation of PV1+ to ground					
0x0025	Insulation of PV2+ to ground					
0x0026	Insulation of PV- to ground					
0x0027	Country code					
0x0028	CT current	int	16			Unit 0.01A, the highest bit is the sign bit, and the negative number means the current flows to the inverter
0x0029	CT power	Int	16			Unit 0.01KW, the highest digit is the sign bit, and the negative number indicates the power flow to the inverter
0x002A	GCFI Valid Value	Uint16	16			Unit 1mA
0x002B	DCI value phase R	Int	16			Unit 1mA
0x002C	DCI value phase S	Int	16			Unit 1mA
0x002D	DCI value phase T	Int	16			Unit 1mA

2.2.3 Read Built-in Combiner Data through address List

Fault List

Byte0

bit	Description	Remarks
Bit0		over voltage Alarm
Bit1		over voltage Alarm
Bit2		over voltage Alarm
Bit3		over voltage Alarm
Bit4		Reversed
Bit5		Reversed
Bit6		Reversed
Bit7		Reversed

Byte1

bit	Description	
Bit0		PV21 over voltage alarm
Bit1		PV22 over voltage alarm
Bit2		PV23 over voltage alarm
Bit3		PV24 over voltage alarm
Bit4		Reversed
Bit5		Reversed
Bit6		Reversed
Bit7		Reversed

Byte2

bit	Description	Remarks
Bit0		Under voltage alarm
Bit1		Under Voltage Alarm
Bit2		Under Voltage Alarm
Bit3		Under Voltage alarm
Bit4		Reversed
Bit5		Reversed
Bit6		Reversed
Bit7		Reversed

Byte3

bit	Description	Remarks
Bit0		PV21 Under Voltage Alarm
Bit1		PV222 Under Voltage Alarm
Bit2		PV23 Under Voltage Alarm
Bit3		PV23 Under Voltage Alarm
Bit4		Reversed
Bit5		Reversed

Bit6		Reversed
Bit7		Reversed

Byte4

bit	Description	Remarks
Bit0		PV11 Reflux Power Fault
Bit1		PV12 Reflux Power Fault
Bit2		PV13 Reflux Power Fault
Bit3		PV14 Reflux Power Fault
Bit4		Reversed
Bit5		Reversed
Bit6		Reversed
Bit7		Reversed

Byte5

bit	Description	Remarks
Bit0		PV21 Reflux Power Fault
Bit1		PV22 Reflux Power Fault
Bit2		PV23 Reflux Power Fault
Bit3		PV21 Reflux Power Fault
Bit4		Reserved
Bit5		Reserved
Bit6		Reserved
Bit7		Reserved

Byte6

bit	Description	Remarks
Bit0		PV11 Over Current Alarm
Bit1		PV12 Over Current Alarm
Bit2		PV13 Over Current Alarm
Bit3		PV14 Over Current Alarm
Bit4		Reserved
Bit5		Reserved
Bit6		Reserved
Bit7		Reserved

Byte7

bi	Description	Remarks
Bit0		PV21 Over Current Alarm
Bit1		PV22 Over Current Alarm
Bit2		PV23 Over Current Alarm
Bit3		PV24 Over Current Alarm
Bit4		Reserved
Bit5		Reserved
Bit6		Reserved
Bit7		Reserved

Byte8

bit	Description	Remarks
Bit0		PV11 FUSE Fault
Bit1		PV12 FUSE Fault
Bit2		PV13 FUSE Fault
Bit3		PV14 FUSE Fault
Bit4		Reserved
Bit5		Reserved
Bit6		Reserved
Bit7		Reserved

Byte9

bit	Description	Remarks
Bit0		PV21 FUSE Fault
Bit1		PV12 FUSE Fault
Bit2		PV13 FUSE Fault
Bit3		PV14 FUSE Fault
Bit4		Reserved
Bit5		Reserved
Bit6		Reserved
Bit7		Reserved

Built-in Combiner Address List

Address	Definition	Variable type	Length	Range	Default Value	Remarks
0x0100	Fault 1	Uint	16			High-Byte:byte1 Low-Byte:byte0
0x0101	Fault 2	Uint	16			High-Byte:byte3 Low-Byte:byte2
0x0102	Fault 3	Uint	16			High-Byte:byte5 Low-Byte:byte4
0x0103	Fault 4	Uint	16			High-Byte:byte7 Low-Byte:byte6
0x0104	Fault 5	Uint	16			High-Byte:byte7 Low-Byte:byte6
PV Input Message						
Addres	Define	Variable Type	Length	Range	Default Value	Remarks
0x0105	PV1 Voltage	Uint	16	0-1000V		Unit 0.1V
0x0106	PV1 Current	Uint	16	-20-20A		Unit 0.01A
0x0107	PV2 Voltage	Uint	16	0-1000V		Unit 0.1V
0x0108	PV2 Current	Uint	16	-20-20A		Unit 0.01A
0x0109	PV3 Voltage	Uint	16	0-1000V		Unit 0.1V
0x010A	PV3 Current	Uint	16	-20-20A		Unit 0.01A
0x010B	PV4 Voltage	Uint	16	0-1000V		Unit 0.1V

0x010C	PV4 Current	Uint	16	-20-20A		Unit 0.01A
0x010D	PV5 Voltage	Uint	16	0-1000V		Unit 0.1V
0x010E	PV5 Current	Uint	16	-20-20A		Unit 0.01A
0x010F	PV6 Voltage	Uint	16	0-1000V		Unit 0.1V
0x0110	PV6 Current	Uint	16	-20-20A		Unit 0.01A
0x0111	PV7 Voltage	Uint	16	0-1000V		Unit 0.1V
0x0112	PV7 Current	Uint	16	-20-20A		Unit 0.01A
0x0113	PV8 Voltage	Uint	16	0-1000V		Unit 0.1V
0x0114	PV8 Current	Uint	16	-20-20A		Unit 0.01A
0x0115 to 0x011F	Reversed					

2.3 Read Production Information(Function code 0x04)

2.3.1 Read the data format

By function code 0x04,query allow for all register,command format as below:

Master request message format：

Slave Address	Function Code	Starting Address	Numbers of register	CRC16
1 byte	1 byte	2 bytes	2 bytes	2 bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Hi Byte Hi Byte

Slave Response Message Format

Slave Address	Function Code	Byte Count	1Register-1 Value	…	Register-N Value	CRC 16
1 byte	1byte	1byte	2bytes	N-2	2 bytes	2 bytes
Byte	Byte	Byte	Hi Byte Lo Byte	…	Hi Byte Lo Byte	Lo Byte Hi Byte

Request example (view running status)：

Request ：

Slave Address	0x01
Function Code	0x04
Register Address(Hi)	0x10
Register Address(Lo)	0x00
NO. of registers(Hi)	0x00
NO. Of register(Lo)	0x01
CRC16(Lo)	0x35

CRC16(Hi）	0x0A

Response：

Slave Address	0x01
Function Code	0x04
Count bytes	0x02
Value of Register(Hi)	0x00
Value of Register (Lo)	0x00
CRC 16(Lo)	0xB9
CRC16(Hi)	0x30

2.3.2 Address Table(Read the setting of inverter)

Address Table (Register Parameter hold by inverter)

Inverter Start-up Parameter						
Address	Definition	Variable Type	Length	Range	Default Value	Remarks
0x1000	Grid connection waiting time			0-1000		unit second
0x1001	Grid power increasing Rate	Uint	16			Percentage of rated power / min
0x1002	Grid connection waiting time after grid fault recovery	Uint	16	0-1000		unit second
0x1003	Grid power increasing after grid fault recovery	Uint	16			Percentage of rated power / min
0x1004	Over voltage protection value before grid connection	Uint	16			Unit 0.1V
0x1005	Under voltage protection value before grid connection	Uint	16			Unit 0.1V
0x1006	Over frequency protection value before grid connection	Uint	16			Unit 0.01Hz
0x1007	Under frequency protection value before grid connection	Uint	16			Unit 0.01Hz
0x1008 to 0x100F	Reversed					
Parameter setting for grid voltage protection						

Address	Definition	Variable Type	Length	Range	Default Value	Remarks
0x1010	Enabled register for grid voltage protection	Uint	16			
0x1011	Class 1 over voltage protection value	Uint	16	10-300		Unit 0.1V
0x1012	Trigger time for class 1 over voltage protection	Uint	16	0-65536		Unit 10ms
0x1013	Class 2 over voltage protection value	Uint	16	10-300		Unit 0.01A
0x1014	Trigger time for class 1 over voltage protection	Uint	16	0-65536		Unit 10ms
0x1015	Class 1 under voltage protection value	Uint	16	10-300		Unit 0.01A
0x1016	Trigger time for class 1 under voltage protection	Uint	16	0-65536		Unit 10ms
0x1017	Class 2 under voltage protection value	Uint	16	10-300		Unit 0.01A
0x1018	Trigger time for class 2 under voltage protection	Uint	16	0-65536		Unit 10ms
0x1019	10-Min Over voltage protection value	Uint	16	10-300		Unit 0.01A
0x101A to 0x101F	Reversed					
Parameter setting for grid frequency protection						
Address	Definition	Variable Type	Length	Range	Default Value	Remarks
0x1020	Enabled register for grid frequency Protection	Uint	16			
0x1021	Class 1 over frequency protection value	Uint	16	50-55		Unit 0.01Hz
0x1022	Trigger Time For class 1over frequency protection	Uint	16	0-65536		Unit 10ms
0x1023	Class 2 over	Uint	16	50-		Unit 0.01Hz

	frequency protection value			55		
0x1024	Trigger Time For class 2 over frequency protection	Uint	16	0-65536		Unit 10ms
0x1025	Class 1 under frequency protection value	Uint	16	45-55		Unit 0.01Hz
0x1026	Trigger Time For class1 under frequency protection	Uint	16	0-65536		Unit 10ms
0x1027	Class 2 under frequency protection value	Uint	16	45-55		Unit 0.01Hz
0x1028	Trigger Time For class 2 under frequency protection	Uint	16	0-65536		Unit 10ms
0x1029 to 0x102F	Reversed					
Parameter setting(Input current DCI protection)						
Address	Definition	Variable Type	Length	Range	Default Value	Remarks
0x1030	Enable register for DCI protection					
0x1031	Class 1 DCI protection value	Uint	16	0-2000		Unit 1mA
0x1032	Trigger Time For class 1 DCI protection	Uint	16	0-65536		Unit 10ms
0x1033	Class 2 DCI protection value	Uint	16	0-2000		Unit 1mA
0x1034	Trigger Time for class 2 DCI protection	Uint	16	0-65536		Unit 10ms
0x1035	DCI injection test value	Uint	16	0-65536		Unit 1mA
Active power & Remote on/off control						
Address	Definition	Variable Type	Length	Range	Default Value	Remarks
0x1040	Enabled register for					Enable

	active& remote on/off control					function remote on/off control only
0x1041	Percentage of active power output			0-1000		Unit 0.1%
0x1042	Remote on/off					55on/66 off
0x1043	Staring point for power derate cause by grid voltage					
0x1044	Ending point for power derate cause by grid voltage					0.1V
0x1045 to 0x104F	Reversed					
Parameter Setting(Active power derate by varying frequency）						
Address	Definition	Variable Type	Length	Range	Default Value	Remarks
0x1050	Enable & Working Mode Selection register					
0x1051	Staring rate for over frequency power derate	Uint	16	0-5500		Unit 0.01Hz
0x1052	Over Frequency power derate rate	Uint	16	0-100		Unit 1%
0x1053	After over frequency power derate restore ,maximum rate for power recovery	Uint	16	0-10000		Unit 0.01Hz
0x1054	After over frequency power derate restore ,minimum rate for power recovery	Uint	16	0-10000		Unit 0.01Hz
0x1055	Waiting time After over frequency power derate restore	Uint	16	0-65535		Unit 0.02s
0x1056	Over frequency recovery rate	Uint	16	0-100		Unit 1%
0x1057	Staring rate for under frequency power derate	Uint	16	0-5500		Unit 0.01Hz
0x1058	Under Frequency	Uint	16	0-100		Unit 1%

	power derate rate					
0x1059	After under frequency power derate restore ,maximum rate for power recovery	Uint	16	0-10000		Unit 0.01Hz
0x105A	Waiting time After under frequency power derate restor	Uint	16	0-65535		Unit 0.02S
0x105B	Under frequency recovery rate	Uint	16	0-100		Unit 1%
Reactive Parameter Setting						
Address	Definition	Variable	Length	Range	Default	Remarks
0x1060	Enable & Working Mode register					
0x1061	Power Factor					Hi byte indicate the symbol and Lo byte indicate the power factor
0x1062	Regular reactive power percentage					Hi byte indicate the percentage
0x1063	P-cos Φ Curve mode first point power factor					
0x1064	P-cos Φ Curve mode first point power percentage					
0x1065	P-cos Φ Curve mode second point power factor					
0x1066	P-cos Φ Curve mode second point power percentage					
0x1067	P-cos Φ Curve mode third point power factor					
0x1068	P-cos Φ Curve mode third point power percentage					

0x1069	P-cos Φ Curve mode fourth point power factor					
0x106A	P-cos Φ Curve mode fourth point power percentage					
0x106B	P-cos Φ curve mode lockin voltage value					
0x106C	P-cos Φ curve mode lockout voltage value					
0x106D	Starting voltage of Q-U curve mode 1 (high voltage)					
0x106E	Ending voltage of Q-U curve mode 1 (high voltage)					
0x106F	Starting voltage of Q-U curve mode 1 (low voltage)					
0x1070	Ending voltage of Q-U curve mode 1 (low voltage)					
0x1071	Lockin power from Q-U curve mode1					
0x1072	lockout power from Q-U curve mode1					
0x1073	Max reactive power of Q-U curve mode1					
0x1074	Response time for reactive power from Q-U curve mode1					
0x1075	Starting voltage of Q-U curve mode2 high voltage					
0x1076	Ending voltage of Q-U curve mode 2 high voltage					
0x1077	starting voltage for Q-U curve mode 2 low voltage					
0x1078	Ending voltage of Q-U curve mode 2 low voltage					
0x1079	Lockin power from Q-U curve mode 2					
0x107A	Lockout power from					

	Q-U curve mode 2					
0x107B	Max reactive power from Q-U curve mode 2					
0x107C	Response time for reactive power from Q-U curve mode2					
0x107D	Reactive phase					
LVRT parameter setting						
Address	Definition	Variable Type	Length	Range	Default value	Remarks
0x1080	LVRT ENABLE REGISTER					
0x1081	Enter the value of LVRT voltage					
0x1082	Voltage for first point (LVRT curve)					
0x1083	Time for first point (LVRT Curve)					
0x1084	Voltage for second point (LVRT curve)					
0x1085	Time for second point (LVRT Curve)					
0x1086	Voltage for third point (LVRT curve)					
0x1087	Time for third point (LVRT Curve)					
0x1088	Voltage for fourth point (LVRT curve)					
0x1089	Time for fourth point (LVRT Curve)					
0x108A	Reactive current coefficient k					
0x108B	Waiting time for voltage recovery					
0x108C	Power recovery rate					
0x1090-0x109F	reserved					
Other safety Parameter setting						
0x10A0	Is-landing enable register					
0x10A1	GFCI enable register					
0x10A2	Isolation Resistor enable register					

0x10A3	Insulation resistance protection value					
Other	reserved					

The definition of the enable register:

Grid voltage protection enable register：

Address：0x1010

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Reversed	Reversed	Reversed	Reversed	Enable bit (class 2 under voltage protection)	Enable bit (class 1 under voltage protection)	Enable bit (class 2 over voltage protection)	Enable bit (class 1 over voltage protection)
Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed

Grid frequency protection enable register

Address：0x1020

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
reversed	Reversed	Reversed	reversed	Enable bit (class 2 under frequency protection)	Enable bit (class 1 under frequency protection)	Enable bit (class 2 over frequency protection)	Enable bit (class 1 over frequency protection)
Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed

(Grid current DCI protection)enable register

Address：0x1030

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Reversed	Reversed	Reversed	Reversed	Reversed	0：disable 1：enable DCI testing function	Enable Bit(class 2 grid DCI protection)	Enable Bit(class 1 grid DCI protection)
Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed

Active power & remote on/off control enable register

Address：0x1040

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Reversed	Reversed	Reversed	Reversed	Reversed	Enable Bit(power derate by grid voltage)	Enable bit(remote on/off control)	Enable bit(power derate by active power)

Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed

Parameter setting(Active power derate by varying frequency)

Address：0x1050

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed		Enable Bit(Active power derate by varying frequency)
Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed

Enable register(Reactive parameter setting)

Address：0x1060

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Reversed	Reversed	Reversed	Reversed	00： Enable Bit(Reactive mode1) 01： Enable Bit(Reactive mode2) 02： Enable Bit(Reactive mode3) 03： Enable Bit(Reactive mode4) 04： Enable Bit(Reactive mode5) 05： Enable Bit(Reactive mode6)			Enable Bit(Reactive Setting）
Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed

Parameter Setting(LVRT Function)：0x1080

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed		0： disable 1： Enable
Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed

Parameter Setting(islanding)

Address：0x10A0

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed		0： disable 1： Enable
Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed

Parameter setting (GFCI function)

Address：0x10A1

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed		0： disable 1： Enable
Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed

Parameter Setting(Insulation Resistance function)

：0x10A2

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Enable PEE testing 0： disable 1： enable	Enable insulation Resistance 0： Disabel 1： enable
Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed

Parameter Setting (PE testing)

Address：0x10A4

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed		0： disable 1： Enable
Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed	Reversed

2.3.3（20-33k, 30-40k, 50-70k）Address Table (Read setting of Built-in Combiner)

Parameter Setting(Built-in Combiner)						
Address	Definitiion	Variable Value	Length	Range	Default Value	Remarks
0x1100	Number of hardware loops	Uint		0-16	6	
0x1101	Value of under voltage Protection	int	16		0x07D0	Unit 0.1v
0x1102	Value of Over Voltage protection	int	16		0X2710	Unit 0.1v
0x1103	Value of reflux power	int	16		0X0064	Unit 0.01A
0x1104	Value of over current protection	int	16		0x04B0	Unit 0.01A
0x1105	Enable over voltage testing	Uint	16		0x0707	
0x1106	Enable under voltage testing	Uint	16		0x0707	
0x1107	Enable reflux power testing	Uint	16		0x0707	
0x1108	Enable over current testing	Uint	16		0x0707	
0x1109	Enable fuse testing	Uint	16		0x0707	

0x110A to 0x110F	Reversed					

2.3.4 Address table(Read production information)

Definition list for barcode ：

ID	Number of digits	Value	Remarks
1	No.1	“S” or “Z” or ‘T’	“S” mean Sofar solar， Other OEM products using other letter to instead
2	No.2、3	“A1” or “B1” or “C1” or “C2” or “C3” or “D1” or “E1” or “F1” ……	“A1”: 1-3KW single-phase one-way inverter， “B1”: 3-5KW single-phase two-way inverter， “C1” :10-20KW three-phase inverter, “C2” :10-20KW three-phase inverter +WIFI， “C3” :10-20KW three-phase inverter +GPRS， “C4” :10-20KW three-phase inverter external， “D1” : 30-40KW three-phase inverter， “D3” :30-40KW three-phase inverter +GPRS， “D4” :30-40KW three-phase inverter external， “E1” :3KW,5KW energy storage system inverter， “F1” :4-12KW three-phase inverter， “F2” :4-12KW three-phase inverter +WIFI， “F3” :4-12KW three-phase inverter +GPRS， “F4” :4-12KW three-phase inverter external， “G1” :30-40KW three-phase inverter(YFL)， “G3” :30-40KW three-phase inverter(YFL)+GPRS， “G4” :30-40KW three-phase inverter(YFL) external， “H1” :3-6KW G2 single-phase two-way inverter， “I1” :50-70KW three-phase inverter(YFL) “J1” :50-70KW three-phase inverter(WK)， “J2” :50-70KW G2 three-phase inverter， “K1” :7.5KW single-phase inverter， “L1” :20-33KW G2 three-phase inverter， “M1” :3-6KW single-phase hybrid inverter， “N1” :10-15KW G2 three-phase inverter， “A3” :1.1~3.3KW G3 single-phase one-way inverter，
3	No.4	C	C: internal
		E	E: Europe、Australia、India、South America…
		U	U: North America
1-6KW Barcode configuration table			
		Barcode	configuration
4	No.5、6	S0	WiFi

		S1	WiFi + DC switch
		S2	DC switch
		S3	None
		S4	GPRS
		S5	DC switch + GPRS
		S6	DC switch + Ethernet
10-20K Barcode configuration table			
4	No.5、6	S0	None
		S1	None
		S2	Standard equipment
		S3	None
		S4	Standard equipment + SPD(LEVEL2, DC)
		S5	Standard equipment + SPD(LEVEL2, DC) + SPD(LEVEL2, AC)
30-40K Barcode configuration table			
4	No.5、6	S0	Standard
		S1	Standard + SPD(DC)
		S2	Standard + SPD(DC)+ SPD(AC)
		S3	Standard + WiFi
		S4	Standard + SPD(DC) + WiFi
		S5	Standard + SPD(DC)+ SPD(AC) + WiFi
50-70KW Barcode configuration table			
4	No.5、6	S1	Standard
		S2	Standard + SPD(LEVEL2, AC)
4-12K Barcode configuration table			
4	No.5、6	S0	Standard equipment
3KW,5KW HYD Barcode configuration table			
4	No.5、6	S2	Standard + GPRS
		S3	Standard + WIFI
3-6K			
4	No.5、6	S0	Standard
		S1	Standard + DC switch
7.5K Single Phase Barcode configuration table			
4	No.5、6	S0	Standard + DC switch
20-33K G2 Barcode configuration table			
4	No.5、6	S0	Standard
3-6KW HYD Barcode configuration table			
4	No.5、6	S0	Standard + DC switch
		S1	Standard + DC switch + WIFI
10-15K G2Barcode configuration table			
4	No.5、6	S0	Standard
1.1~3.3K-G3Barcode configuration table			
4	No.5、6	S0	Standard

		S1	Standard + DC switch
5	No.7、8	Inverter related power	Please read the related power description below
6	No.9	Year（2000-2035）	2000（0）,2001（1）,2002（2）,2003（3）,2004（4）,2005（5）,2006（6）,2007（7）,2008（8）,2009（9）,2010（A）,2011（B）,2012（C）,2013（D）,2014（E）,2015（F）,2016（G）,2017（H）,2018（J）,2019（K）,2020（L）,2021（M）,2022（N）,2023（P）,2024（Q）,2025（R）,2026（S）,2027（T）,2028（U）,2029（V）,2030（W）,2031（X）,2032（Y）,2033（Z）
7	No.10	Month	January（1）,February（2）,March（3）,April（4）,May（5）,June（6）,July（7）,August（8）,September（9）,October（A）,November（B）,December（C）
8	No.11	Day	1st（1）,2nd（2）,3rd（3）,4th（4）
			5th（5）,6th（6）,7th（7）,8th（8）
			9th（9）,10th（A）,11th（B）,12th（C）
			13th（D）,14th（E）,15th（F）,16th（G）
			17th（H）,18th（J）,19th（K）,20th（L）
			21st（M）,22nd（N）,23rd（P）,24th（Q）
			25th（R）,26th（S）,27th（T）,28th（U）
			29th（V）,30th（W）,31st（X）
9	No.12、13、14	xxx	Number of unit produce that day

Related power description (NO. 7&8 digits of barcode)

1. 1-3KW Single phase single MPPT inverter (“A”)

11:1100W Model: SOFAR 1100TL 16:1600W Model：SOFAR 1600TL

22:2200W Model: SOFAR 2200TL 27:2700W Model: SOFAR2700TL

30:3000W Model: SOFAR 3000TL

2. 3-6KW Single phase Dual MPPTS inverter (“B”)

30:3000W Model: SOFAR3000TLM 36:3680W Model: SOFAR3680TLM

40:4000W Model:SOFAR4000TLM 46:4600W Model: SOFAR4600TLM

50:5000W Model:SOFAR5000TLM

3、10-20KW Three Phase Inverter Dual MPPTS(“C”)

10:10000W Model：SOFAR 10000TL 15:15000W Model：SOFAR 15000TL

17:17000W Model：SOFAR 17000TL 20:20000W Model：SOFAR 20000TL

4、30-40KW Three Phase Inverter Dual MPPTS(“D”)

30:30000W Model：SOFAR 30000TL 33:33000W Model：SOFAR 33000TL

36:36000W Model：SOFAR 36000TL 40:40000W Model：SOFAR 40000TL

5、3-6KW Single Phase Battery charge& Discharge control inverter(“E”)

30:3000W Model：ME3000 SP

50:5000W Model：ME5000 SP

6、4-12KW Three Phase Inverter Dual MPPTS(“F”)

04:4000W Model：SOFAR 4KTL-X 05:5000W Model：SOFAR 5KTL-X

06:6000W Model：SOFAR 6KTL-X 08:8000W Model：SOFAR 8KTL-X

10:10000W Model：SOFAR 10KTL-X 12:12000W Model：SOFAR 12KTL-X

7、30-40KW Three Phase Inverter Dual MPPTS IGBT: infineon(“G”)

30:30000W Model：SOFAR 30000TL 33:33000W Model：SOFAR 33000TL

36:36000W Model：SOFAR 36000TL 40:40000W Model：SOFAR 40000TL

8、3-6KW Single phase Dual MPPTS -G2(“H”)

30:3000W Model：SOFAR 3KTLM-G2 36:3680W Model：SOFAR 3.6KTLM-G2

40:4000W Model：SOFAR 4KTLM-G2 46:4600W Model：SOFAR 4.6KTLM-G2

50:5000W Model：SOFAR 5KTLM-G2 60:6000W Model：SOFAR 6KTLM-G2

9、50-70KW Three Phase Inverter Three MPPTS IGBT: infineon(“G”)(Reversed, no produce yet)

50:50000W Model：SOFAR 50000TL 60:60000W Model：SOFAR 60000TL

70:70000W Model：SOFAR 70000TL

10、50-70KW Three Phase Inverter Three MPPTS(“J”)

50:50000W Model：SOFAR 50000TL 60:60000W Model：SOFAR 60000TL

70:70000W Model：SOFAR 70000TL

11、7.5KW Single Phase Dual MPPTS(“K”)

75:7500W Model：SOFAR 7.5KTLM-G2

12、20-33KW Three Phase Dual MPPTS(“L”)

20:20000W Model：SOFAR 20000TL-G2 25:25000W Model：SOFAR 25000TL-G2

30:30000W Model：SOFAR 30000TL-G2 33:33000W Model：SOFAR 33000TL-G2

Related Information about inverter’s production(Independent space 16 bytes)						
Address	Definition	Variable Type)	Length (byte)	Range	Default Value	Remarks
0x2000	Production Code	uint	2			**Single Phase1-3k：** 0x11：1.1KW 0x16：1.6KW 0x22：2.1KW 0x27：2.7KW 0x30：3KW **Single Phase 3-6k：**

						0x30：3KW 0x36：3.6KW 0x40：4KW 0x46：4.6KW 0x50：5KW 0x60：6KW **Single Phase 3-6k-G2：** 0x30：3KW 0x36：3.6KW 0x40：4KW 0x46：4.6KW 0x50：5KW 0x60：6KW **Single phase 7.5k：** 0x75:7.5KW **Three Phase：** 0：5KW 1：6KW 2：8KW 3：10KW 4：15KW 5：17KW 6：20KW 7：25KW 8：30KW 9：33KW 10：40KW 11：36KW
0x2001 to 0x2007	Series Number	string	14			please see the definition list for series number
0x2008 to0x2009	Software version	string	4			Software version format:Vxxxx For example：V020, V021
0x200Ato 0x200B	Hardware Version	string	4			Hardware version format:Vxxxx For example：V020, V021
0x200C to 0x200F	Reversed					

（20-33k, 30-40k, 50-70k） Related Information about built in combiner’s production(Independent space 16 bytes)

Address	Definition	Variable Type	Length	Range	Default Value	Remarks
0x2100 to	Hardware					

0x2101	version					
0x2102 to 0x2103	Software version					
0x2104 To 0x210F	Reversed					

2.3.5 Address Table(Read calibration data measure by inverter)

Measurement Calibration for inverter						
Address	Definition	Variable Value	length	Range	Default Value	Remarks
0x3000	ratio correction factor for VPV1			0.95-1.05		
0x3001	deviation value for VPV1			+-15V		
0x3002	Ratio correction factor for IPV1			0.95-1.05		
0x3003	Deviation value for IPV1			+-1A		
0x3004	Ratio correction factor for PV1 power			0.95-1.05		
0x3005	Deviation value for PV1 power			+-100W		
0x3006	Ratio Correction factor for VPV2			0.95-1.05		
0x3007	Deviation value for VPV2			+-15V		
0x3008	Ratio Correction factor forIPV2			0.95-1.05		
0x3009	Deviation value for IPV2			+-1A		
0x300A	Ratio correction factor for PV2 power			0.95-1.05		
0x300B	Deviation value for PV2 power			+-100W		
0x300C	Ratio correction factor for Vbus			0.95-1.05		
0x300D	Deviation value for Vbus			+-15V		
0x300E	Ratio Correction factor of output			0.95-1.05		

	power					
0x30F0	Deviation value for output power			+-100W		
0x3010	Ratio Correction factor for R phase voltage			0.95-1.05		
0x3011	Deviation value for R phase voltage			+-15V		
0x3012	Ratio Correction factor for R phase current			0.95-1.05		
0x3013	Deviation value for R phase current			+-1A		
0x3014	Ratio Correction factor for S phase voltage			0.95-1.05		
0x3015	Deviation value for S phase voltage			+-15V		
0x3016	Ratio Correction factor for S phase current			0.95-1.05		
0x3017	Deviation value for S phase current			+-1A		
0x3018	Ratio Correction factor for T phase voltage			0.95-1.05		
0x3019	Deviation value for T phase voltage			+-15V		
0x301A	Ratio Correction factor for T phase current			0.95-1.05		
0x301B	Deviation value for T phase current			+-1A		
0x301C	Ratio Correction factor for sub CPU VPV1			0.95-1.05		
0x301D	Deviation value for sub CPU VPV1			+-15V		
0x301E	Ratio Correction factor for sub CPU VPV2			0.95-1.05		
0x301F	Deviation value for sub CPU VPV2			+-15V		
0x3020 to 0x302F	Reversed					

Parameter Setting for factory Mode						
0x3100	Register for factory Mode					Low byte:Factory mode Byte7: 1 enable 0 disable Byte6 ：Clear calibration factor1，0 Byte5,4,3 1 Enable 0 Disable Byte1：1PV2 working 0 not working Byte0,1 working 0 not working byte: percentage of output current ,128 mean 100%

2.4 Writing Parameter(Function Code0x13)

2.4.1 Write the Data Frame

Master Request message format

Slave Address	Function Code	Starting Address	Numbers of the register	Register-1 value	···	Register-N value	CRC16
1 byte	1 byte	2 bytes	2byte	2bytes	N-2	2bytes	2 bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	…	Hi Byte Lo Byte	Hi Byte Lo Byte

Slave response message format

Slave Address	Function Code	Starting Address	Numbers of the register	Register-1 value	···	Register-N value	CRC16
1 byte	1 byte	2 bytes	2byte	2bytes	N-2	2bytes	2 bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	…	Hi Byte Lo Byte	Hi Byte Lo Byte

Request frame example (query running status)：

Request：

Slave Address	0x01
Function Code	0x13
Register Address Hi	0x10
Register Address Lo	0x00
NO. Of register Hi	0x00
No. Of register Lo	0x01
Register Value(Hi)	0x00
Register value(Lo)	0x01
CRC 16(Lo)	0x31
CRC16(Hi)	0x96

Response：

Slave Address	0x01
Function Code	0x13
Register Address Hi	0x10
Register Address Lo	0x00
NO. Of register Hi	0x00
No. Of register Lo	0x01
Register-1 Value(Hi)	0x00
Register-1 value(Lo)	0x01
CRC 16(Lo)	0x31
CRC16(Hi)	0x96

2.4.2 Address Table(Write the setting for inverter)

Start-up parameter setting						
Address	Defintion	Variable Type	Length	Range	Default Vale	Remarks
0x1000	Grid connection waiting time			0-1000		Unit second
0x1001	Grid power increasing Rate	Uint	16			Percentage of rated power / min
0x1002	Grid connection waiting time after grid fault recovery	Uint	16	0-1000		Unit second
0x1003	Grid power increasing after grid fault recovery	Uint	16			Percentage of rated power / min
0x1004	Over voltage protection value before grid connection	Uint	16			Unit 0.1V

0x1005	Under voltage protection value before grid connection	Uint	16			Unit 0.1V
0x1006	Over frequency protection value before grid connection	Uint	16			Unit 0.01Hz
0x1007	Under frequency protection value before grid connection	Uint	16			Unit 0.01Hz
0x1008 to 0x100F	Reversed					
Parameter setting for grid voltage protection						
Address	Definition	Variable type	Length	Range	Default Value	Remarks
0x1010	Enabled register for grid voltage protection	Uint	16			
0x1011	Class 1 over voltage protection value	Uint	16	10-300		Unit 0.1V
0x1012	Trigger time for class 1 over voltage protection	Uint	16	0-65536		Unit 10ms
0x1013	Class 2 over voltage protection value	Uint	16	10-300		Unit 0.01A
0x1014	Trigger time for class 1 over voltage protection	Uint	16	0-65536		Unit 10ms
0x1015	Class 1 under voltage protection value	Uint	16	10-300		Unit 0.01A
0x1016	Trigger time for class 1 under voltage protection	Uint	16	0-65536		Unit 10ms
0x1017	Class 2 under voltage	Uint	16	10-300		Unit 0.01A

	protection value					
0x1019	Trigger time for class 2 under voltage protection	Uint	16	0-65536		Unit 10ms
0x1019	10-Min Over voltage protection value	Uint	16	10-300		Unit 0.01A
0x101A to 0x101F	Reversed					
Parameter setting for grid frequency protection						
Address	Definition	Variable Type	Length	Range	Default Value	Remarks
0x1020	Enabled register for grid frequency Protection	Uint	16			
0x1021	Class 1 over frequency protection value	Uint	16	50-55		Unit 0.01Hz
0x1022	Trigger Time For class 1over frequency protection	Uint	16	0-65536		Unit 10ms
0x1023	Class 2 over frequency protection value	Uint	16	50-55		Unit 0.01Hz
0x1024	Trigger Time For class 2 over frequency protection	Uint	16	0-65536		Unit 10ms
0x1025	Class 1 under frequency protection value	Uint	16	45-55		Unit 0.01Hz
0x1026	Trigger Time For class1 under frequency protection	Uint	16	0-65536		Unit 10ms
0x1027	Class 2 under frequency protection value	Uint	16	45-55		Unit 0.01Hz
0x1028	Trigger Time For	Uint	16	0-65536		Unit 10ms

	class 2 under frequency protection					
0x1029 to 0x102F	Reversed					
Parameter Setting (input current DCI protection)						
Address	Definition	Variable type	Length	Range	Default Value	Remarks
0x1030	Enable register for DCI protection					
0x1031	Class 1 DCI protection value	Uint	16	0-2000		Unit mA
0x1032	Trigger Time For class 1 DCI protection	Uint	16	0-65536		Unit 10ms
0x1033	Class 2 DCI protection value	Uint	16	0-2000		Unit 1mA
0x1034	Trigger Time for class 2 DCI protection	Uint	16	0-65536		Unit 10ms
0x1035	DCI injection test value	Uint	16	0-65536		Unit 1mA
Active power & Remote on/off control						
Address	Definition	Variable Type	Length	Range	Default Value	Remarks
0x1040	Enabled register for active& remote on/off control					Enable function Remote on/off control only
0x1041	Percentage of active power output			0-1000		0.1%
0x1042	Remote on/off					
0x1043	Staring point for power derate cause by grid voltage					Unit 0.1V
0x1044	Ending point for power derate cause by grid voltage					Unit 0.1V
0x1045 to0 0x104F	Reversed					
Parameter Setting(Active power derate by varying frequency						
Address	Definition	Variable	Length	Range	Default	Remarks

		type			Value	
0x1050	Enable & Working Mode register					
0x1051	Staring rate for over frequency power derate	Uint	16	0-5500		Unit 0.01Hz
0x1052	over Frequency power derate rate	Uint	16	0-100		Unit 1%
0x1053	After over frequency power derate restore ,maximum rate for power recovery	Uint	16	0-10000		Unit 0.01Hz
0x1054	After over frequency power derate restore ,minimum rate for power recovery	Uint	16	0-10000		Unit 0.01Hz
0x1055	Waiting time After over frequency power derate restore	Uint	16	0-65535		Unit0.02s
0x1056	Over frequency recovery rate	Uint	16	0-100		Unit 1%
0x1057	Staring rate for under frequency power derate	Uint	16	0-5500		Unit 0.01Hz
0x1058	Under Frequency power derate rate	Uint	16	0-100		Unit 1%
0x1059	After under frequency power derate restore ,maximum rate for power recovery	Uint	16	0-10000		Unit 0.01Hz
0x105A	Waiting time After under frequency power derate	Uint	16	0-65535		Unit 0.02s

	restor					
0x105B	Under frequency recovery rate	Uint	16	0-100		Unit1%
Reactive Parameter Setting						
Address	Definition	Variable	Length	Range	Default Value	Remarks
0x1060	Enable & Working Mode register					
0x1061	Power Factor					Hi Byte indicate the symbol and Lo byte indicate the power factor
0x1062	Regular reactive power percentage					Hi byte indicate the percentage
0x1063	P-cos Φ Curve mode first point power factor					
0x1064	P-cos Φ Curve mode first point power percentage					
0x1065	P-cos Φ Curve mode second point power factor					
0x1066	P-cos Φ Curve mode second point power percentage					
0x1067	P-cos Φ Curve mode third point power factor					
0x1068	P-cos Φ Curve mode third point power percentage					
0x1069	P-cos Φ Curve mode fourth point power factor					
0x106A	P-cos Φ Curve mode fourth					

	point power percentage					
0x106B	P-cos Φ curve mode lockin voltage value					
0x106C	P-cos Φ curve mode lockout voltage value					
0x106D	Starting voltage of Q-U curve mode 1 (high voltage)					
0x106E	Ending voltage of Q-U curve mode 1 (high voltage)					
0x106F	Starting voltage of Q-U curve mode 1 (low voltage)					
0x1070	Ending voltage of Q-U curve mode 1 (low voltage)					
0x1071	Lockin power from Q-U curve mode1					
0x1072	Lockout power from Q-U curve mode1					
0x1073	Max reactive power of Q-U curve mode1					
0x1074	Response time for reactive power from Q-U curve mode1					
0x1075	Starting voltage of Q-U curve mode2 high voltage					
0x1076	Ending voltage of Q-U curve mode 2 high voltage					
0x1077	Starting voltage for Q-U curve mode 2 low					

	voltage					
0x1078	Ending voltage of Q-U curve mode 2 low voltage					
0x1079	Lockin power from Q-U curve mode 2					
0x107A	Lockout power from Q-U curve mode 2					
0x107B	Q-max reactive power from Q-U curve mode 2					
0x107C	Response time for reactive power from Q-U curve mode2					
0x107D	Reactive phase					
LVRT Parameter setting						
Address	Definition	Variable Type	Length	Range	Default Value	Remarks
0x1080	LVRT ENABLE REGISTER					
0x1081	Enter the value of LVRT voltage					
0x1082	Voltage for first point (LVRT curve)					
0x1083	Time for first point (LVRT Curve)					
0x1084	Voltage for second point (LVRT curve)					
0x1085	Time for second point (LVRT Curve)					
0x1086	Voltage for third point (LVRT curve)					
0x1087	Time for third point (LVRT Curve)					
0x1088	Voltage for fourth point					

	(LVRT curve)					
0x1089	Time for fourth point (LVRT Curve)					
0x108A	Reactive current coefficient k					
0x108B	Waiting time for voltage recovery					
0x108C	Power recovery rate					
Other safety Parameter setting						
0x10A0	Is-landing enable register					
0x10A1	GFCI enable register					
0x10A2	Isolation Resistor enable register					
0x10A3	Insulation resistance protection value					
Parameter Setting (Factory Mode)						
0x3100	Register for factory mode					low byte:Factory mode Byte7: 1 enable 0 disable Byte6 ： Clear calibration factor1，0 Byte5,4,3 1 Enable 0 Disable Byte1 ： 1 PV2 working 0 not working Byte0, 1 working 0 not working High byte: percentage of

						output current ,128 mean 100%
Other	Reversed					

2.4.3 Address Table (write setting for Built-in combiner 20-33k, 30-40k, 50-70k）

Parameter Setting(Built-in combiner)						
Address	Definition	Variable Type	Length	Range	Default Value	Remarks
0x1100	Number of current loops					
0x1101	Value of under voltage Protection	Uint	16			Hi byte byte1; Lo byte, byte0
0x1102	Value of Over Voltage protection	Uint	16			Hi byte byte3; Lo byte, byte2
0x1103	Value of reflux power	Uint	16			Hi byte byte5; Lo byte, byte4
0x1104	Value of over current protection	Uint	16		0x04B0	Unit 0.01A
0x1105	enable over voltage testing	Uint	16			
0x1106	enable under voltage testing	Uint	16			
0x1107	enable reflux power testing	Uint	16			
0x1108	Enable over current testing	Uint	16			
0x1109	Enable fuse testing	Uint	16			
0x1109 to 0x110F	Reversed					

2.5 Write related production information to inverter(Function Code 0x21)

2.5.1 Format of write-in data

Master request message Format：

Slave Address	Function Code	Starting Address	Numbers Of registers	Register-1 Value	…	Register(N) value	CRC16
1byte	1byte	2 bytes	2 bytes	2 bytes	N-2	2 bytes	2 bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	…	Hi Byte Lo Byte	Hi Byte Lo Byte

Slave response message format：

Slave Address	Function Code	Starting Address	Numbers Of registers	Register-1 Value	…	Register(N) value	CRC16
1byte	1byte	2 bytes	2 bytes	2 bytes	N-2	2 bytes	2 bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	…	Hi Byte Lo Byte	Hi Byte Lo Byte

Request frame example (query running status)：

Request：

Slave Address	0x01
Function Code	0x21
Register Address Hi	0x20
Register Address Lo	0x00
NO. Of register Hi	0x00
No. Of register Lo	0x01
Register Value(Hi)	0x00
Register value(Lo)	0x00
CRC 16(Lo)	0XE6
CRC16(Hi)	0x65

Response：

Slave Address	0x01
Function Code	0x21
Register Address Hi	0x20
Register Address Lo	0x00
NO. Of register Hi	0x00
No. Of register Lo	0x01
Register Value(Hi)	0x00
Register value(Lo)	0x00
CRC 16(Lo)	0XE6
CRC16(Hi)	0x65

2.5.2 Write Related production information to inverter address Table

（20-33k, 30-40k, 50-70k） Related Information about built in combiner's production(Independent space 16 bytes)（20-33k, 30-40k, 50-70k）						
Address	Definition	Variable Type	Length	Range	Default Value	Remarks
0x2000	Production Code					(Read-only)
0x2001 To 0x2006	Series Number					
0x2007 to 0x2008	Software version					(Read-only)
0x2009 to 0x200A	Hardware version					
0x200B to 0x200F						

2.6 Measurement calibration write-in (Only for SOFAR use)

2.7 Maintenance related information (Only for SOFAR use)

2.8 Read (EEPROM) History energy and Event list(function code 0x50)

Through function code to search the Data information of allowed registers. The command format show as follows:

2.8.1 Read Data Format

Master request message format:

Slave addreess	Function code	Starting address	Register information	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Lo Byte Hi Byte

Slave response message format：

Slave addreess	Function code	Number of Byte	Valid Data	CRC16
1 Byte	1 Byte	2 Bytes	N Bytes	2 Bytes
Byte	Byte	Byte	N Bytes	Lo Byte Hi Byte

2.8.2 Read History energy and Address list of events

Address	Definition	Variable type	Length(bytes)	Range	Default	Remarks
0x6000	Today's generation	Hex	2*24			Register Information Bar have no practical meaning. It is 0.
0x6001	Daily generation of this month	Hex	2*31			Register Information Bar have no practical meaning. It is 0.
0x6002	Daily generation of this month	Hex	4*12			Register Information Bar have no practical meaning. It is 0.
0x6003	Total generation those year	Hex	4			
0x6004	Power Generation of specific Year N	Hex	4 or 4*20			The value of Register is N, for example, N=2 means it is the energy of recent specific Year 2. If N=0xFFFF, read all energy of recent 20 years.
06005	Recent Nth event record	Hex	8			The value of register is N, for example, N=2 means it is the Recent Second Event Record. Total record recent 100 event lists
0x6006	Recent Nth Timing Record	Hex	12			The value of register is N, for example, N=2 means it is the Recent Second Timing Record. Total record recent 10 Timing records.
0x6007	Recent clear Power to Zero Records	HEX	6			The value of register is N, for example, N=2 means it is the Recent Second Timing Record. Totally records recent 10 Timing records.
0x6008	Recent clear Events to Zero Records	HEX	6			The value of register is N, for example, N=2 means it is the Second Clear events Record. Totally records recent 10 Clear events records.

The valid data format (event record response data frame)

ID number of event and the time when event was sent							
ID number of Event	Year YY	Month MM	Day DD	Weak	Hour HH	Minute MM	Second SS

The valid data format (timing record response data frame)

Time before auto adjustment						Time after auto adjustment					
Second ss	Minute mm	Hour Hh	Day DD	Month MM	Year YY	Second ss	Minute mm	Hour Hh	Day DD	Month MM	Year YY

The valid data format (Energy clearing record response data frame)

Energy clearing time					
Second ss	Minute mm	Hour hh	Day DD	Month MM	Year YY

The valid data format (Event list clearing record response data frame)

Events clearing time					
Second ss	Minute mm	Hour hh	Day DD	Month MM	Year YY

2.9 Write (EEPROM) history energy(function code 0x51)

2.9.1 Write Data Foramt

Master request message format:

Slave addree ss	Function code	Starting address	Register information	Number of bytes written	Data to be written	CRC16
1 byte	1 byte	2 bytes	1 byte	1 byte	N bytes	2 bytes
Byte	Byte	Hi Byte Lo Byte	Byte	Byte	N Bytes	Hi Byte Lo Byte

Slave response message format：

Slave addree ss	Function code	Starting address	Register information	Number of bytes written	Data to be written	CRC16
1 byte	1 byte	2 bytes	1 byte	1 byte	N bytes	2 bytes
Byte	Byte	Hi Byte Lo Byte	Byte	Byte	N Bytes	Hi Byte Lo Byte

2.9.2 Write history energy

Address	Definition	Variable type	(bytes) Length(by tes)	Range	Defa ults	Remarks
0x6000	Today's Generation	Hex	2*24			
0x6001	Daily generation of this month	Hex	2*31			
0x6002	Daily generation of this month	Hex	4*12			
0x6003	Total generation those year	Hex	4			
0x6004	Genearation of recent specific Year N	Hex	4 or 4*20			The value of register is N, for example, N=2 means it is the energy of recent specific Year 2. If N=0xFFFF, read all energy of recent 20 years.

2.10 Read History energy stored by SD card(function code 0x60)

Through function code 0x60 to search data information of allowed registers, the command format show as follows:

2.10.1 Read Data Format

Master request message format:

Slave addreess	Function code	Starting address	Register information	CRC16
1 Byte	1 Byte	2 Bytes	3bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	3 Byte	Lo Byte Hi Byte

Slave response message format：

Slave addreess	Function code	Number of Bytes	Valid Data	CRC16
1 Byte	1 Byte	1 Byte	N Bytes	2 Bytes
Byte	Byte	Byte	N Bytes	Lo Byte Hi Byte

2.10.2 Read history energy

Address	Definition	Variable type	(bytes) Length(by	Range	Defa ults	Remarks

			tes)			
0x7000	One day's energy	Hex	2*24			
0x7001	Daily energy of this month	Hex	2*31			
0x7002	Monthly energy of this year	Hex	4*12			

The data format of the information bar of reading one day's total energy from SD card is as follows:

Year YY	Month MM	Day DD
BCD code	BCD code	BCD code

The data format of the information bar of reading one month's total energy from SD card is as follows:

Year YY	Month MM	Reserved
BCD code	BCD code	00

The data format of the information bar of reading one year's total energy from SD card is as follows:

Year YY	Reserved	Reserved
BCD code	00	00

2.11 Read time(function code 0x10)

Through function code 0x10 to read time, the command format is as follows:

2.11.1 Read Data Format

Master request message format:

Slave addreess	Function code	Starting address	Number of Registers	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	0x00 0x00	Lo Byte Hi Byte

Slave response message format：

Slave addreess	Function code	Number of Byte	Valid parameters	CRC16
1 Byte	1 Byte	2 Bytes	N Bytes	2 Bytes
Byte	Byte	Byte	N Bytes	Lo Byte Hi Byte

2.11.2 The Address of reading time

Address	Definition	Variable type	(bytes) Length(bytes)	Range	Defaults	Remarks

0x8000	System time	BCD	7			
0x8001	Power-on time on control board today	BCD	7			

The valid data format of the response frame is as follows:

Second ss	Minute mm	Hour hh	Weak D	Day DD	Month MM	Year YY

2.12 Return to factory setting(function code 0x30)

2.12.1 Read Data Format

Through function code 0x30 to return to factory setting

Master request message format:

Slave addreess	Function code	Starting address	Number of Registers	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Lo Byte Hi Byte

Slave response message format：

Slave addreess	Function code	Starting address	Number of Registers	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Lo Byte Hi Byte

2.13 Clear today’s energy(function code 0x31)

2.13.1 Read Data Format

Through function code 0x31 to clear today’s energy

Master request message format:

Slave addreess	Function code	Starting address	Number of Registers	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Lo Byte Hi Byte

Slave response message format：

Slave addreess	Function code	Starting address	Number of Registers	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Lo Byte Hi Byte

2.14 Return to default values of current Country code setting(function code 0x32)

2.14.1 Data Format

Through function code 0x32 to return to default values of current Country code setting

Master request message format:

Slave addreess	Function code	Starting address	Number of Registers	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Lo Byte Hi Byte

Slave response message format：

Slave addreess	Function code	Starting address	Number of Registers	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Lo Byte Hi Byte

2.15 Clear energy(function code 0x33)

2.15.1 Data Format

Through function code 0x33 to clear energy

Master request message format:

Slave addreess	Function code	Starting address	Number of Registers	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Lo Byte Hi Byte

Slave response message format：

Slave addreess	Function code	Starting address	Number of Registers	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Lo Byte Hi Byte

2.16 Clear Eventlist(function code 0x34)

2.16.1 Data Format

Through function code to clear eventlist

Master request message format:

Slave addreess	Function code	Starting address	Number of Registers	CRC16

1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Lo Byte Hi Byte

Slave response message format：

Slave addreess	Function code	Starting address	Number of Registers	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Lo Byte Hi Byte

2.16 Read control Byte of relay(function code 0x35)

2.16.1 Data Format

Through function code 0x35 to read control Byte of relay

Master request message format:

Slave addreess	Function code	Starting address	Number of Registers	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	0x00 0x00	0x00 0x00	Lo Byte Hi Byte

Slave response message format：

Slave addreess	Function code	Number of Bytes	Control Byte	CRC16
1 Byte	1 Byte	1 Byte	1 Byte	2 Bytes
Byte	Byte	Byte	Byte	Lo Byte Hi Byte

Note: The control Byte of relay is defined as follows

0x00	Production
0x01	Alarm
0x02	Alarm(configable)

2.17 Set control Byte of this relay(function code 0x36)

2.17.1 Data Format

Through function code 0x36 to set control Byte of this relay

Post request message format:

Slave addreess	Function code	Starting address	Reserved Byte	Control Byte	CRC16
1 Byte	1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes
Byte	Byte	0x00 0x00	0x00	Byte	Lo Byte Hi Byte

Note: The control Byte of relay define as follows

0x00	Production
0x01	Alarm
0x02	Alarm(configable)
0XAA	Relay disable(Disable the control function of this relay)

Slave response message format：

Slave address	Function code	Starting address	Control Byte	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	0x00 0x00	Hi Byte Lo Byte	Lo Byte Hi Byte

2.18 Read the Alarm ID number when control Byte is Alarm(configable)(function code 0x37)

2.18.1 Data Format

Through function code 0x37 to read the Alarm ID number when control Byte is Alarm(configable)

Master request message format:

Slave addreess	Function code	Starting address	Number of Registers	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	0x00 0x00	0x00 0x00	Lo Byte Hi Byte

Slave response message format：

Slave addreess	Function code	Number of Byte	Control Byte	CRC16
1 Byte	1 Byte	1 Byte	N Bytes	2 Bytes
Byte	Byte	Byte	N Bytes	Lo Byte Hi Byte

2.19 Set the Alarm ID number when control Byte is Alarm(configable)(function code 0x38)

2.19.1 Data Format

Through function code 0x38 to set the Alarm ID number when control Byte is Alarm(configable)

Master request message format:

Slave addreess	Function code	Starting address	Number of bytes written	ID number to be written	CRC16
1 Byte	1 Byte	2 Bytes	1 Byte	N Bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	Byte	N Bytes	Hi Byte Lo Byte

Slave response message format：

Slave addreess	Function code	Starting address	Number of bytes written	ID number to be written	CRC16
1 Byte	1 Byte	2 Bytes	1 Byte	N Bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	Byte	N Bytes	Hi Byte Lo Byte

2.20 Read test Flags(function code 0x45)

2.20.1 Data Format

Through function code 0x45 to read test Flags

Master request message format:

Slave addreess	Function code	Starting address	Number of Registers	CRC16
1 Byte	2 Bytes	2 Bytes	2 Bytes	2 Bytes
Byte	0x45	0x00 0x00	0x00 0x00	Lo Byte Hi Byte

Slave response message format：

Slave addreess	Function code	Number of Byte	Control Byte	CRC16
1 Byte	1 Byte	1 Byte	1 Byte	2 Bytes
Byte	Byte	Byte	1 Bytes	Lo Byte Hi Byte

Flag Definition:

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Reserved	Reserved	Reserved	Reserved	Reserved	T2 Test Completion Flag	Aging Test Completion Flag	T1 Test Completion Flag

2.21 Set test flags(function code 0x46)

2.21.1 Data Format

Through function code 0x46 to set test flags

Master request message format:

Slave addreess	Function code	Starting address	Parameters to be written	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Hi Byte Lo Byte

Slave response message format：

Slave addreess	Function code	Starting address	Parameters to be written	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	Byte	Hi Byte Lo Byte	Hi Byte Lo Byte	Hi Byte Lo Byte

2.22 Read Parameters of MPPT scan (function ocde 0x49)

2.22.1 Data Format

Through function code 0x49, read parameters of MPPT scan

Master request message format:

Slave addreess	Function code	Number of Registers	Parameters to be written	CRC16
1 Byte	1 Byte	2 Bytes	5 Bytes	2 Bytes
Byte	0x49	0x00 0x05		Hi Byte Lo Byte

Read voltage of PV1: RPV1V
Read voltage of PV2: RPV2V
Read current of PV1: RPV1I
Read current of PV2: RPV2I

Slave response message format:

Slave addreess	Function code	Number of bytes	Read values	CRC16
1 Byte	1 Byte	1 Byte	200 Bytes	2 Bytes
Byte	0x49	200		Lo Byte Hi Byte

2.23 Read Parameters of MPPT scan setting (function ocde 0x52)

2.23.1 Data Format

Through function code 0x52, read the parameters of MPPT scan setting

Master request message format:

Slave addreess	Function code	Starting address	Number of Registers	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	0x52	0x00 0x00	0x00 0x00	Lo Byte Hi Byte

Slave response message format:

Slave addreess	Function code	Number of bytes	Enable	Scan Frequency	CRC16
1 Byte	1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes
Byte	Byte	Byte	1 Bytes	1 Bytes	Lo Byte Hi Byte

Enable byte 0x55: Enable scanning 0XAA:Disable scanning

Scan Frequency: 1-60 minutes

2.24 Set MPPT scan parameters(function code 0x53)

2.24.1 Data Format

Through function code 0x53 to set MPPT scan parameters.

Post request message format:

Slave addreess	Function code	Starting address	Enable control byte	Scan frequency	CRC16
1 Byte	1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes
Byte	0x53	0x00 0x00	1 Byte	1 Byte	Lo Byte Hi Byte

Slave response message format:

Slave addreess	Function code	Starting address	Enable control byte	Scan frequency	CRC16
1 Byte	1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes
Byte	0x53	0x00 0x00	1 Byte	1 Byte	Lo Byte Hi Byte

Scanning enable control byte 0x55: Enable scanning 0XAA:Disable scanning

Scan Frequency: 1-60 minutes

2.25 MPPT scan test command(function code 0x54)

2.25.1 Data Format

Through function code 0x54 to do MPPT scan test.

Master request message format:

Slave addreess		Function code	Starting address	Test control byte	CRC16
1 Byte		1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte		0x54	0x00 0x00	0x00 0x55	Lo Byte Hi Byte

Slave response message format:

Slave addreess	Function code	Starting address	Test control byte	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	0x54	0x00 0x00	0x00 0x55	Lo Byte Hi Byte

When you read I-V curve and P-V curve on website, need to send 2.22 command. Waiting the correct response then send 2.10 command.

2.26 Power real-time control command(function code 0x06)

2.26.1 Data Format

Through 0x06 function code, limit output power in real-time

Master request message format:

Slave addreess	Function code	Starting address	Power control byte	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	0x06	0x90 0x00		Lo Byte Hi Byte

Slave response message format:

Slave addreess	Function code	Starting address	Power control byte	CRC16
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes
Byte	0x06	0x90 0x00		Lo Byte Hi Byte

Power control byte is the percentage of output power, if the percentage is 100%, then the power control byte should be 0x00 0x64